

Task - 2

Problem 2.A

Case ①

let exactly k contribute & $n-k$ not contribute.

Then, good is provided.

If contributor switches to not contributing → outcome changes from
no advantage in switching

if non-contributor switches to contributing
no advantage in switching → outcome changes from

$2 \rightarrow 3$
worse

∴ Nash equilibrium. → exactly k contribute.

$1 \rightarrow 2$
worse.

Case ② $k > 0$

let $< k$ contribute } no public good provided.

If a contributor switches to not contributing → Not Nash

outcome

$4 \rightarrow 3$
better

Case ③

let $> k$ contribute } public good provided.

If a contributor switches to not contrib... →

Not Nash

$2 \rightarrow 1$
better

Case ④ :- 0 contribute }

if anyone contributes → $3 \rightarrow 4$
worse

Nash

Nash equilibrium is when exactly k contribute & $n-k$ don't.

or 0 contribute

Problem 2. B.

Case

① a portion of stag is always preferred over hare.

② Then let's say $\geq m$ ^{$f < n$} are catching stag.

Then if one catching hare goes to catch stag,
payoff increases for him $\geq$
Not Nash.

③ if $< m$, then, hunter catching stag will prefer to catch hare rather than getting nothing.
Not Nash

④ if all hunt stag,
all prefer hunting stag to hare Nash
All hunt stag

⑤ If all hunt hare,
switching to stag gives nothing.
Nash
All hunt hare.

∴ Nash Equilibria are

① All hunt stag

② All hunt hare

Case 2 if $> k$ hunters ($m \leq k \leq n$) hunters hunt stag, they would rather prefer hare.

(A) Let $> k$ hunters go for stag, they would prefer hare. (Not Nash).

(B) Let exactly k hunters go for stag
→ stag hunters don't switch since they (Nash) prefer $1/k$ of stag over hare
→ hare hunters don't switch because they prefer hare over $\frac{1}{k+1}$ of stag

(C) If a hunters, ($m \leq a < k$) go for stag, hare hunters would prefer to switch (Not Nash)

(D) If $< m$ hunters go for stag, stag hunters would prefer to switch (Not Nash)

(E) All hunt hare.
no point in hunting stag

Nash
All hunt hare

∴ Nash Equilibria are

① All hunt hare

② Exactly k hunt stag

Task 3

Initially,

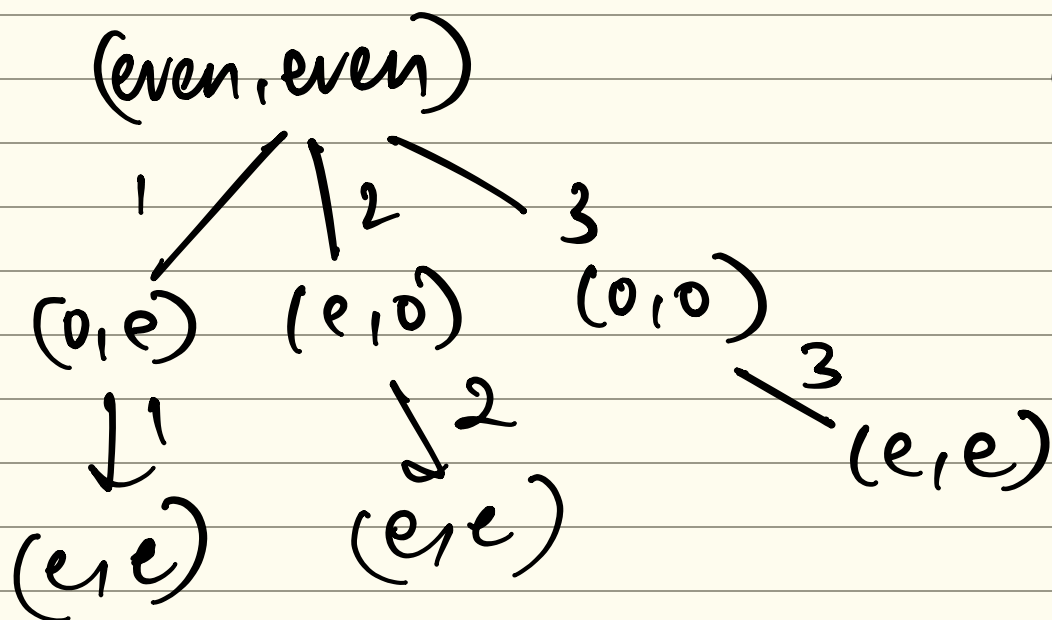
10
A

10
B.

1 → take from A

2 → take from B

3 → take from A & B.



Can always be made (e, e) no matter what opp. plays.

& since opp can't get (e, e) if we keep making it (e, e) before opp turn,

This is winning strat as $(0, 0)$ is (e, e) .

We can always win.

∴ The strategy is to always keep both piles have even no. of stones.